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Structured Query Language

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Marks 1 10 questions

Q1. GATE CSE 2020

Open solution

Consider a relational database containing the following schemas.

Catalogue

sno	pno	cost
S1	P1	150
S1	P2	50
S1	P3	100
S2	P4	200
S2	P5	250
S3	P1	250
S3	P2	150
S3	P5	300
S3	P4	250

Suppliers

sno	sname	location
S1	M/s Royal furniture	Delhi
S2	M/s Balaji furniture	Bangalore
S3	M/s Premium furniture	Chennai

Parts

pno	pname	part_spec
P1	Table	Wood
P2	Chair	Wood
P3	Table	Steel
P4	Almirah	Steel
P5	Almirah	Wood

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The primary key of each table is indicated by underlying the constituent fields.

```
SELECT s.sno, s.sname

FROM Suppliers s, Catalogue c

WHERE s.sno = c.sno AND

      Cost > (SELECT AVG (cost)

                  FROM Catalogue

                  WHERE pno = 'P4'

                  GROUP BY pno);
```

The number of rows returned by the above SQL query is

(A) 4 (B) 5 (C) 2 (D) 0

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Q2. GATE CSE 2018

Open solution

Consider the following two tables and four queries in SQL.

Book (isbn, bname), Stock (isbn, copies)

Query 1:

```
SELECT B.isbn, S.copies
FROM Book B INNER JOIN Stock S
ON B.isbn = S.isbn;
```

Query 2:

```
SELECT B.isbn, S.copies
FROM Book B LEFT OUTER JOIN Stock S
ON B.isbn = S.isbn;
```

Query 3:

```
SELECT B.isbn, S.copies
FROM Book B RIGHT OUTER JOIN Stock S
ON B.isbn = S.isbn;
```

Query 4:

```
SELECT B.isbn, S.copies
FROM Book B FULL OUTER JOIN Stock S
ON B.isbn = S.isbn;
```

Which one of the queries above is certain to have an output that is a superset of the outputs of the other three queries?

(A) Query 1 (B) Query 2 (C) Query 3 (D) Query 4

Q3. GATE CSE 2017 Set 2

Open solution

Consider the following tables T1 and T2.

T1

P	Q
2	2
3	8
7	3
5	8
6	9
8	5
9	8

T2

R	S
2	2
8	3
3	2
9	7
5	7
7	2

In table T1, **P** is the primary key and **Q** is the foreign key referencing **R** in table T2 with on-delete cascade and on-update cascade. In table T2, **R** is the primary key and **S** is the foreign key referencing **P** in table T1 with on-delete set NULL and on-update cascade. In order to delete record $\langle 3, 8 \rangle$ from table T1, the number of additional records that need to be deleted from table T1 is _____.

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Q4. GATE CSE 2017 Set 1

Open solution

Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName). An instance of the schema EMP and a SQL query on it are given below.

EMP

EmpId	EmpName	DeptName
1	XYA	AA
2	XYB	AA
3	XYC	AA
4	XYD	AA
5	XYE	AB
6	XYF	AB
7	XYG	AB
8	XYH	AC
9	XYI	AC
10	XYJ	AC
11	XYK	AD
12	XYL	AD
13	XYM	AE

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
      (SELECT DeptName, COUNT(EmpId) AS
                        EC(DeptName, Num)
       FROM EMP
       GROUP BY DeptName)
```

The output of executing the SQL query is _____.

Q5. GATE CSE 2015 Set 3

Open solution

Consider the following relation

Cinema(theater, address, capacity)

Which of the following options will be needed at the end of the SQL query

SELECT $P1$.address
FROM Cinema $P1$

such that it always finds the addresses of theaters with maximum capacity?

(A) WHERE $P1$.capacity >= All (select $P2$.capacity from Cinema $P2$)

(B) WHERE $P1$.capacity >= Any (select $P2$.capacity from Cinema $P2$)

(C) WHERE $P1$.capacity > All (select max($P2$.capacity) from Cinema $P2$)

(D) WHERE $P1$.capacity > Any (select max($P2$.capacity) from Cinema $P2$)

Q6. GATE CSE 2015 Set 1

Open solution

SELECT operation in SQL is equivalent to

(A) the selection operation in relational algebra

(B) the selection operation in relational algebra, except that SELECT in SQL retains duplicates

(C) the projection operation in relational algebra

(D) the projection operation in relational algebra, except that SELECT in SQL retains duplicates

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Q7. GATE CSE 2014 Set 1

Open solution

Given the following statements:
S1: A foreign key declaration can always be replaced by an equivalent check assertion in SQL.

S2: Given the table R(a,b,c) where a and b together form the primary key, the following is a valid table definition.

```
CREATE TABLE S (
 a INTEGER,
 d INTEGER,
 e INTEGER,
 PRIMARY KEY (d),
 FOREIGN KEY (a) references R)
```

Which one of the following statements is **CORRECT**?

(A) S1 is TRUE and S2 is FALSE.
(B) Both S1 and S2 are TRUE.
(C) S1 is FALSE and S2 is TRUE.
(D) Both S1 and S2 are FALSE.

Q8. GATE CSE 2012

Open solution

Which of the following statements are TRUE about an SQL query?
P: An SQL query can contain a HAVING clause even if it does not have a GROUP BY clause
Q: An SQL query can contain a HAVING clause only if it has a GROUP BY clause
R: All attributes used in the GROUP BY clause must appear in the SELECT clause
S: Not all attributes used in the GROUP BY clause need to appear in the SELECT clause

(A) P and R
(B) P and S
(C) Q and R
(D) Q and S

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Q9. GATE CSE 2010

Open solution

A relational schema for a train reservation database is given below.

Passenger(pid, pname, age)
Reservation(pid, class, tid)

Table: Passenger

pid	pname	Age
0	'Sachin'	65
1	'Rahul'	66
2	'Sourav'	67
3	'Anil'	69

Table: Reservation

pid	class	tid
0	'AC'	8200
1	'AC'	8201
2	'SC'	8201
5	'AC'	8203
1	'SC'	8204
3	'AC'	8202

What pids are returned by the following SQL query for the above instance of the tables?

```
SELECT pid
FROM   Reservation
WHERE  class = 'AC' AND
       EXISTS (SELECT *
               FROM Passenger
               WHERE age > 65 AND
               Passenger.pid = Reservation.pid)
```

(A) 1, 0 (B) 1, 2 (C) 1, 3 (D) 1, 5

Q10. GATE CSE 2004

Open solution

Consider the following relation schema pertaining to a students database:

Students (rollno, name, address)
Enroll(rollno,courseno, coursename)
Where the primary keys are shown underlined. The number of tuples in the student and Enroll tables are 120 and 8 respectively. What are the maximum and minimum number of tuples that can be present in (Student * Enroll), where '*' denotes natural join?

(A) 8, 8 (B) 120, 8 (C) 960, 8 (D) 960, 120

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Marks 2 34 questions

Q11. GATE CSE 2025 Set 2

Open solution

Consider the following relational schema:

Students (rollno : integer, name: string, age: integer, cgpa: real)

Courses (courseno : integer, cname: string, credits: integer)

Enrolled (rollno : integer, courseno : integer, grade: string)

Which of the following options is/are correct SQL query/queries to retrieve the names of the students enrolled in course number (i.e., courseno) 1470?

(A)
```
SELECT S.name
FROM Students    S
WHERE EXISTS  (SELECT * FROM Enrolled E
               WHERE E.courseno = 1470
               AND E.rollno = S.rollno);
```

(B)
```
SELECT S.name
FROM Students    S
WHERE SIZEOF  (SELECT * FROM Enrolled E
               WHERE E.courseno = 1470
                     AND E.rollno = S.rollno) > 0;
```

(C)
```
SELECT S.name
FROM Students    S
          WHERE 0 < (SELECT COUNT(*)
          FROM Enrolled E
          WHERE E.courseno = 1470
                AND E.rollno = S.rollno);
```

(D)
```
SELECT S.name
FROM Students S NATURAL JOIN Enrolled E
WHERE E.courseno = 1470;
```

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Q12. GATE CSE 2025 Set 1

Open solution

Consider the following database tables of a sports league.

player(pid, pname, age)

team(tid, tname, city, cid)

coach(cid, cname)

members(pid, tid)

An instance of the table and an SQL query are given.

player

pid	pname	age
1	Jasprit	31
2	Atharva	24
3	Ishan	26
4	Axar	30

coach

cid	cname
101	Ricky
102	Mark
104	Teevor

team

tid	tname	city	cid
10	MI	Mumbai	102
20	DC	Delhi	101
30	PK	Mohali	103

members

pid	tid
1	10
2	30
3	10
4	20

```
SELECT MIN(P.age)
FROM player P
WHERE P.pid IN (
    SELECT M.pid
    FROM team T, coach C, members M
    WHERE C.cname = 'Mark'
        AND T.cid = C.cid
        AND M.tid = T.tid
)
```

The value returned by the given SQL query is _____ . (Answer in integer)

Q13. GATE CSE 2023

Open solution

Consider the following table named Student in a relational database. The primary key of this table is rollNum.

Student

rollNum	name	gender	marks
1	Naman	M	62
2	Aliya	F	70
3	Aliya	F	80
4	James	M	82
5	Swati	F	65

The SQL query below is executed on this database.

```
SELECT *
FROM Student
WHERE gender = 'F' AND
marks > 65;
```

The number of rows returned by the query is _____.

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Q14. GATE CSE 2022

Open solution

Consider the relational database with the following four schemas and their respective instances.

Student($\underline{sNo}$, sName, dNo) Dept($\underline{dNo}$, dName)

Course($\underline{cNo}$, cName, dNo) Register($\underline{sNo}$, $\underline{cNo}$)

Student

sNo	sName	Dno
S01	James	D01
S02	Rocky	D02
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Dept

dNo	dName
D01	CSE
D02	EEE

Course

cNo	sName	dNo
C11	DS	D01
C12	OS	D01
C21	DE	D02
C22	PT	D02
C23	CV	D03

Register

sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

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SQL Query :

SELECT * FROM Student AS S WHERE NOT EXIST

(SELECT cNo FROM Course WHERE dNo = "D01"

EXCEPT

SELECT cNo FROM Register WHERE sNo = S.sNo)

The number of rows returned by the above SQL query is _____.

Q15. GATE CSE 2021 Set 2

Open solution

The relation scheme given below is used to store information about the employees of a company, where **empId** is the key and **deptId** indicates the department to which the employee is assigned. Each employee is assigned to exactly one department.

emp(empId, name, gender, salary, deptId)

Consider the following SQL query:

```
select deptId, count(*)
from emp
where gender = "female" and salary > (select avg(salary) from emp)
group by deptId;
```

The above query gives, for each department in the company, the number of female employees whose salary is greater than the average salary of

(A) female employees in the department.

(B) employees in the department.

(C) employees in the company.

(D) female employees in the company.

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Q16. GATE CSE 2019

Open solution

A relational database contains two tables Student and Performance as shown below :

Student

Roll_no.	Student_name
1	Amit
2	Priya
3	Vinit
4	Rohan
5	Smita

Performance

Roll_no.	Subject_code	Marks
1	A	86
1	B	95
1	C	90
2	A	89
2	C	92
3	C	80

The primary key of the Student table is Roll_no. For the Performance table, the columns Roll_no. and Subject_code together from the primary key. Consider the SQL query given below:

```
SELECT S.Student_name, sum (P.Marks)
       FROM Student S, Performance P
       WHERE P.Marks > 84
       GROUP BY S.Student_name;
```

The number of rows returned by the above SQL query is _____.

Q17. GATE CSE 2017 Set 2

Open solution

Consider the following database table named *top_scorer*.

player	country	goals
Klose	Germany	16
Ronaldo	Brazil	15
G Müller	Germany	14
Fontaine	France	13
Pelé	Brazil	12
Klinsmann	Germany	11
Kocsis	Hungary	11
Batistuta	Argentina	10
Cubillas	Peru	10
Lato	Poland	10
Lineker	England	10
T Müller	Germany	10
Rahn	Germany	10

Consider the following SQL query:

```
SELECT ta.player FROM top_scorer AS ta
WHERE ta.goals >ALL (SELECT tb.goals
             FROM top_scorer AS tb
             WHERE tb.country = 'Spain')
AND ta.goals >ANY (SELECT tc.goals
             FROM top_scorer AS tc
             WHERE tc.country = 'Germany')
```

The number of tuples returned by the above SQL query is _____.

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Q18. GATE CSE 2016 Set 2

Open solution

Consider the following database table named *water_schemes* :

scheme_no	district_name	capacity
1	Ajmer	20
1	Bikaner	10
2	Bikaner	10
3	Bikaner	20
1	Churu	10
2	Churu	20
1	Dungargarh	10

The number of tuples returned by the following SQL query is _____ .

```
with total(name, capacity) as
    select district_name, sum(capacity)
    from water_schemes
    group by district_name
with total_avg(capacity) as
    select avg(capacity)
    from total
select name
    from total, total_avg
    where total.capacity ≥ total_avg.capacity
```

Q19. GATE CSE 2015 Set 1

Open solution

Consider the following relations:

Student

Roll_No	Student_Name
1	Raj
2	Rohit
3	Raj

Performance

Roll_No	Course	Marks
1	Math	80
1	English	70
2	Math	75
3	English	80
2	Physics	65
3	Math	80

Consider the following SQL query.

```
SELECT S.Student_Name, sum(P.Marks)
FROM Student S, Performance P
WHERE S.Roll_No = P.Roll_No
GROUP BY S.Student_Name
```

The number of rows that will be returned by the SQL query is _____.

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Q20. GATE CSE 2014 Set 1

Open solution

Given the following schema:

```
employees(emp-id, first-name, last-name, hire-date,
          dept-id, salary)
departments(dept-id, dept-name, manager-id, location-id)
```

You want to display the last names and hire dates of all latest hires in their respective departments in the location ID 1700. You issue the following query:

```
SQL>SELECT last-name, hire-date
    FROM employees
    WHERE (dept-id, hire-date) IN
    (SELECT dept-id, MAX(hire-date)
    FROM employees JOIN departments USING(dept-id)
    WHERE location-id = 1700
    GROUP BY dept-id);
```

What is the outcome?

(A) It executes but does not give the correct result.

(B) It executes and gives the correct result.

(C) It generates an error because of pairwise comparison.

(D) It generates an error because the GROUP BY clause cannot be used with table joins in a sub-query.

Q21. GATE CSE 2014 Set 2

Open solution

SQL allows duplicate tuples in relations, and correspondingly defines the multiplicity of tuples in the result of joins. Which one of the following queries always gives the same answer as the nested query shown below:

```
Select * from R where a in (select S.a from S)
```

(A)
```
Select R.* from R, S where R.a = S.a
```

(B)
```
Select distinct R.* from R, S where R.a = S.a
```

(C)
```
Select R.* from R, (select distinct a from S) as S1 where R.a = S1.a
```

(D)
```
Select R.* from R, S where R.a = S.a and is unique R
```

Q22. GATE CSE 2014 Set 3

Open solution

Consider the following relational schema:

employee(empId,empName,empDept)
customer(custId,custName,salesRepId,rating)

salesRepId is a foreign key referring to **empId** of the employee relation. Assume that each employee makes a sale to at least one customer. What does the following query return?

```
SELECT empName
FROM employee E
WHERE NOT EXISTS (SELECT custId
                  FROM customer C
                  WHERE C.salesRepId = E.empId
                  AND C.rating <> 'GOOD');
```

(A) Names of all the employees with at least one of their customers having a 'GOOD' rating.

(B) Names of all the employees with at most one of their customers having a 'GOOD' rating.

(C) Names of all the employees with none of their customers having a 'GOOD' rating.

(D) Names of all the employees with all their customers having a 'GOOD' rating.

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Q23. GATE CSE 2013

Open solution

Consider the following relational schema.

Students(rollno: integer, sname: string)
Courses(courseno: integer, cname: string)
Registration(rollno: integer, courseno: integer, percent: real)

Which of the following queries are equivalent to this query in English?

"Find the distinct names of all students who score more than 90% in the course numbered 107"

(I)

```
SELECT DISTINCT S.sname
FROM Students as S, Registration as R
WHERE R.rollno=S.rollno AND R.courseno=107 AND R.percent >90
```

(II) $\Pi_{sname}\left(\sigma_{courseno=107 \wedge percent>90}\left(Registration \bowtie Students\right)\right)$

(III) $\{T \mid \exists S \in Students, \exists R \in Registration\left(S.rollno = R.rollno \wedge R.courseno = 107 \wedge R.percent > 90 \wedge T.sname = S.sname\right)\}$

(IV) $\{\langle S_N \rangle \mid \exists S_R \exists R_P\left(\langle S_R, S_N \rangle \in Students \wedge \langle S_R, 107, R_P \rangle \in Registration \wedge R_P > 90\right)\}$

(A) I, II, III and IV
(B) I, II and III only
(C) I, II and IV only
(D) II, III and IV only

Q24. GATE CSE 2012

Open solution

Consider the following relations A, B and C:

A

Id	Name	Age
12	Arun	60
15	Shreya	24
99	Rohit	11

B

Id	Name	Age
15	Shreya	24
25	Hari	40
98	Rohit	20
99	Rohit	11

C

Id	Phone	Area
10	2200	02
99	2100	01

How many tuples does the result of the following SQL query contain?

```
SELECT A.Id
FROM A
WHERE A.Age > ALL (SELECT B.Age
                   FROM B
                   WHERE B.Name = 'Arun')
```

(A) 4 (B) 3 (C) 0 (D) 1

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Q25. GATE CSE 2011

Open solution

Database table by name `Loan_Records` is given below.

Borrower	Bank_Manager	Loan_Amount
Ramesh	Sunderajan	10000.00
Suresh	Ramgopal	5000.00
Mahesh	Sunderajan	7000.00

What is the output of the following SQL query?

```
SELECT count(*)
FROM (
        (SELECT Borrower, Bank_Manager FROM Loan_Records) AS S
        NATURAL JOIN
        (SELECT Bank_Manager, Loan_Amount FROM Loan_Records) AS T
);
```

(A) 3 (B) 9 (C) 5 (D) 6

Q26. GATE CSE 2011

Open solution

Consider a database table T containing two columns X and Y each of type integer. After the creation of the table, one record (X = 1, Y = 1) is inserted in the table.

Let MX and MY denote the respective maximum values of X and Y among all records in the table at any point in time. Using MX and MY, new records are inserted in the table 128 times with X and Y values being MX + 1, 2*MY + 1 respectively. It may be noted that each time after the insertion, values of MX and MY change.

What will be the output of the following SQL query after the steps mentioned above are carried out?

```
SELECT Y FROM T WHERE X=7;
```

(A) 127 (B) 255 (C) 129 (D) 257

Q27. GATE CSE 2009

Open solution

Consider the following relational schema:

Suppliers(sid: integer, sname: string, city: string, street: string)

Parts(pid: integer, pname: string, color: string)

Catalog(sid: integer, pid: integer, cost: real)

Consider the following relational query on the above database:

```
SELECT S.sname
FROM   Suppliers S
WHERE  S.sid NOT IN ( SELECT C.sid
                      FROM   Catalog C
                      WHERE  C.pid NOT IN ( SELECT P.pid
                                            FROM   Parts P
                                            WHERE  P.color <> 'blue'))
```

Assume that relations corresponding to the above schema are not empty. Which one of the following is the correct interpretation of the above query?

(A) Find the names of all suppliers who have supplied a non-blue part.

(B) Find the names of all suppliers who have not supplied a non-blue part.

(C) Find the names of all suppliers who have supplied only blue parts.

(D) Find the names of all suppliers who have not supplied only blue parts.

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Q28. GATE CSE 2007

Open solution

Consider the table **employee**(empId, name, department, salary) and the two queries Q_1, Q_2 below. Assuming that department 5 has more than one employee, and we want to find the employees who get higher salary than anyone in the department 5, which one of the statements is **TRUE** for any arbitrary employee table?

Q_1:

```
Select e.empId
From employee e
Where not exists
    (Select * From employee s Where s.department = "5" and s.salary >= e.salary)
```

Q_2:

```
Select e.empId
From employee e
Where e.salary > Any
    (Select distinct salary From employee s Where s.department = "5")
```

(A) Q_1 is the correct query.

(B) Q_2 is the correct query.

(C) Both Q_1 and Q_2 produce the same answer.

(D) Neither Q_1 nor Q_2 is the correct query.

Q29. GATE CSE 2006

Open solution

Consider the relation **account (customer, balance)** where customer is a primary key and there are no null values. We would like to rank customers according to decreasing balance. The customer with the largest balance gets rank 1. Ties are not broke but ranks are skipped: if exactly two customers have the largest balance they each get rank 1 and rank 2 is not assigned.

```
Query1:
  Select A.customer, count(B.customer)
  From account A, account B
  Where A.balance <=B.balance
  Group by A.customer

Query2:
  Select A.customer, 1+count(B.customer)
  From account A, account B
  Where A.balance < B.balance
  Group by A.customer
```

Consider these statements about Query1 and Query2.

1. Query1 will produce the same row set as Query2 for some but not all databases.
2. Both Query1 and Query2 are correct implementation of the specification
3. Query1 is a correct implementation of the specification but Query2 is not
4. Neither Query1 nor Query2 is a correct implementation of the specification
5. Assigning rank with a pure relational query takes less time than scanning in decreasing balance order assigning ranks using ODBC.

Which two of the above statements are correct?

(A) 2 and 5

(B) 1 and 3

(C) 1 and 4

(D) 3 and 5

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Q30. GATE CSE 2006

Open solution

Consider the relation "**enrolled (student, course)**" in which (student, course) is the primary key, and the relation "**paid (student, amount)**" where student is the primary key. Assume no null values and no foreign keys or integrity constraints. Given the following four queries:

Query 1:

```
Select student
from enrolled
where student in (select student from paid)
```

Query 2:

```
Select student
from paid
where student in (select student from enrolled)
```

Query 3:

```
Select E.student
from enrolled E, paid P
where E.student = P.student
```

Query 4:

```
Select student
from paid
where exists (Select *
              from enrolled
              where enrolled.student = paid.student)
```

Which one of the following statements is correct?

(A) All queries return identical row sets for any database

(B) Query 2 and Query 4 return identical row sets for all databases but there exist databases for which Query 1 and Query 2 return different row sets.

(C) There exist databases for which Query 3 returns strictly fewer rows than Query 2

(D) There exist databases for which Query 4 will encounter an integrity violation at runtime.

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Q31. GATE CSE 2006

Open solution

Consider a database with three relation instances shown below. The primary keys for the **Drivers** and **Cars** relation are Did and cid respectively and the records are stored in ascending order of these primary keys as given in the tables. No indexing is available in the database.

D: Drivers Relation

Did	Dname	rating	Age
22	Karthikeyan	7	25
29	Salman	1	33
31	Boris	8	55
32	Amoldt	8	25
58	Schumacher	10	35
64	Sachin	7	35
71	Senna	10	16
74	Sachin	9	35
85	Rahul	3	25
95	Ralph	3	53

R: Reserves Relation

Did	cid	Day
22	101	10/10/06
22	102	10/10/06
22	103	8/10/06
22	104	7/10/06
31	102	10/11/06
31	103	6/11/06
31	104	12/11/06
64	101	5/9/06
64	102	8/9/06
74	103	8/9/06

C: Cars relation

cid	Cname	Color
101	Renault	Blue
102	Renault	Red
103	Ferrari	Green
104	Jaguar	Red

What is the output of the following SQL query?

```
Select D.dname
From Drivers D
Where D.did in (SELECT R.did
                From Cars C,Reserves R
                WHERE R.cid = C.cid and C.color = 'red'
                INTERSECT
                SELECT R.did
                From Cars C,Reserves R
                WHERE R.cid = C.cid and C.color = 'green')
```

(A) Karthikeyan, Boris

(B) Sachin, Salman

(C) Karthikeyan, Boris, Sachin

(D) Schumacher, Senna

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Q32. GATE CSE 2006

Open solution

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```
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From Drivers D
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                From Cars C,Reserves R
                WHERE R.cid = C.cid and C.color = 'red'
                INTERSECT
                SELECT R.did
                From Cars C,Reserves R
                WHERE R.cid = C.cid and C.color = 'green')
```

Let n be the number of comparisons performed when the above SQL query is optimally executed. If linear search is used to locate a tuple in a relation using primary key, then n lies in the range

(A) 36 - 40 (B) 44 - 48 (C) 60 - 64 (D) 100 - 104

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Q33. GATE CSE 2005

Open solution

The relation book (title, price) contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list?

```
Select title
 From book as B
 Where (select count(*)
   From book as T
   Where T.price > B.price) < 5;
```

(A) Titles of the four most expensive books

(B) Title of the fifth most inexpensive book

(C) Title of the fifth most expensive book

(D) Titles of the five most expensive books

Q34. GATE CSE 2005

Open solution

A company maintains records of sales made by its salespersons and pays them commission based on each individual's total sales made in a year. This data is maintained in a table with following schema:

salesinfo = (salespersonid, totalsales, commission)

In a certain year, due to better business results, the company decides to further reward its salespersons by enhancing the commission paid to them as per the following formula:

If commission < = 50000, enhance it by 2%
If 50000 < commission < = 100000, enhance it by 4%
If commission > 100000, enhance it by 6%

The IT staff has written three different SQL scripts to calculate enhancement for each slab, each of these scripts is to run as a separate transaction as follows:

```
T1: Update salesinfo
 Set commission = commission * 1.02
 Where commission < = 50000;

T2: Update salesinfo
 Set commission = commission * 1.04
 Where commission > 50000 and commission is < = 100000;

T3: Update salesinfo
 Set commission = commission * 1.06
 Where commission > 100000;
```

Which of the following options of running these transactions will update the commission of all salespersons correctly?

(A) Execute T1 followed by T2 followed by T3

(B) Execute T2, followed by T3; T1 running concurrently throughout

(C) Execute T3 followed by T2; T1 running concurrently throughout

(D) Execute T3 followed by T2 followed by T1

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Q35. GATE CSE 2005

Open solution

In an inventory management system implemented at a trading corporation, there are several tables designed to hold all the information. Amongst these, the following two tables hold information on which items are supplied by which suppliers, and which warehouse keeps which items along with the stock-level of these items.
Supply = (supplierid, itemcode)
Inventory = (itemcode, warehouse, stocklevel)

For a specific information required by the management, following SQL query has been written

```
Select distinct STMP.supplierid
From Supply as STMP
Where not unique (Select ITMP.supplierid
    From Inventory, Supply as ITMP
    Where STMP.supplierid = ITMP.supplierid
    And ITMP.itemcode = Inventory.itemcode
    And Inventory.warehouse = 'Nagpur');
```

For the warehouse at Nagpur, this query will find all suppliers who

(A) do not supply any item
(B) supply exactly one item
(C) supply one or more items
(D) supply two or more items

Q36. GATE CSE 2004

Open solution

The employee information in a company is stored in the relation

Employee (name, sex, salary, deptName)
Consider the following SQL query

```
Select deptName
From Employee
Where sex = 'M'
Group by deptName
Having avg(salary) >
(Select avg (salary) From Employee);
```

It returns the names of the department in which

(A) the average salary is more than the average salary in the company
(B) the average salary of male employees is more than the average salary of all male employees in the company
(C) the average salary of male employees is more than the average salary of employees in the same department.
(D) the average salary of male employees is more than the average salary in the company

Q37. GATE CSE 2004

Open solution

A relational database contains two tables **student** and **department** in which student table has columns **roll_no, name** and **dept_id** and department table has columns **dept_id** and **dept_name**. The following insert statements were executed successfully to populate the empty tables:

```
Insert into department values (1, 'Mathematics')
Insert into department values (2, 'Physics')
Insert into student values (1, 'Navin',1)
Insert into student values (2, 'Mukesh',2)
Insert into student values (3, 'Gita',1)
```

How many rows and columns will be retrieved by the following SQL statement?

```
Select * from student, department;
```

(A) 0 row and 4 columns
(B) 3 rows and 4 columns
(C) 3 rows and 5 columns
(D) 6 rows and 5 columns

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Q38. GATE CSE 2004

Open solution

Consider two tables in a relational database with columns and rows as follows:

Table: Student

Roll_no	Name	Dept_id
1	ABC	1
2	DEF	1
3	GHI	2
4	JKL	3

Table: Department

Dept_id	Dept_name
1	A
2	B
3	C

Roll_no is the primary key of the Student table, Dept_id is the primary key of the Department table and Studetn.Dept_id is a foreign key from
Department. Dept_id.

What will happen if we try to execute the following two SQL statements?

```
(i) update Student set Dept_id = Null where Roll_no =1
(ii) update Department set Dept_id = Null where Dept_id =1
```

(A) Both (i) and (ii) will fail

(B) (i) will fail but (ii) will succeed

(C) (i) will succeed but (ii) will fail

(D) Both (i) and (ii) will succeed

Q39. GATE CSE 2003

Open solution

Consider the set of relations shown below and the SQL query that follows.
Students: (Roll_number, Name, Date_of_birth)
Courses: (Course number, Course_name, Instructor)
Grades: (Roll_number, Course_number, Grade)

```
Select distinct Name
From Students, Courses, Grades
Where Students.Roll_number = Grades.Roll_number
      and Courses.Instructor = 'Korth'
      and Courses.Course_number = Grades.Course_number
      and Grades.grade = 'A';
```

Which of the following sets is computed by the above query?

(A) Names of students who have got an A grade in all courses taught by Korth

(B) Names of students who have got an A grade in all courses

(C) Names of students who have got an A grade in at least one of the courses taught by Korth

(D) None of the above

Q40. GATE CSE 2000

Open solution

In SQL, relations can contain null values, and comparisons with null values are treated as unknown. Suppose all comparisons with a null value are treated as false. Which of the following pairs is not equivalent?

(A) $x = 5$ not $(not\ (x = 5))$

(B) $x = 5$ $x > 4$ and $x < 6$, where x is an integer

(C) $x \neq 5$ not $(x = 5)$

(D) None of the above

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Q41. GATE CSE 2000

Open solution

Given relations r(w, x) and s(y, z), the result of

```
select distinct w,x
from r, s;
```

is guaranteed to be same as r, provided

(A) r has no duplicates and s is non-empty

(B) r and s have no duplicates

(C) s has no duplicates and r is non-empty

(D) r and s have the same number of tuples

Q42. GATE CSE 1999

Open solution

Which of the following is/are correct?

(A) An SQL query automatically eliminates duplicates.

(B) An SQL query will not work if there are no indexes on the relations.

(C) SQL permits attribute names to be repeated in the same relation.

(D) None of the above.

Q43. GATE CSE 1999

Open solution

Consider the set of relations
EMP (Employee-no, Dept-no, Employee-name, Salary)
DEPT (Dept-no, Dept-name, Location)

Write an SQL query to:

(a) Find all employee names who work in departments located at "Calcutta" and whose salary is greater than Rs.50,000.

(b) Calculate, for each department number, the number of employees with a salary greater than Rs.1,00,000.

Q44. GATE CSE 1998

Open solution

Suppose we have a database consisting of the following three relations.

FREQUENTS (student, parlor) giving the parlors each student visits.

SERVES (parlor, ice-cream) indicating what kind of ice-creams each parlor serves.

LIKES (student, ice-cream) indicating what ice-creams each student likes.

(Assume that each student likes at least one ice-cream and frequents at least one parlor)
Express the following in SQL:
Print the students that frequent at least one parlor that serves some ice-cream that they like.

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Answer Key

Marks 1

1. **A**	2. **D**	3. **0**	4. **2.6**	5. **A**
6. **D**	7. **D**	8. **C**	9. **C**	10. **A**

Marks 2

11. **A, C, D**	12. **26**	13. **2**	14. **2**	15. **C**
16. **5**	17. **7**	18. **2**	19. **2**	20. **B**
21. **C**	22. **D**	23. **A**	24. **B**	25. **C**
26. **A**	27. **A**	28. **A**	29. **C**	30. **A**
31. **A**	32. **A**	33. **D**	34. **D**	35. **D**
36. **D**	37. **B**	38. **C**	39. **C**	40. **C**

41. **A**	42. **D**	43. **(a)**
		`SELECT Employee-name`
		`FROM EMP, DEPT`
		`WHERE EMP.Dept-no = DEPT.Dept-no`
		`and EMP.Salary > 50,000 and`
		`DEPT.Location="Calcutta";`
		(b)
		`SELECT count(Employee-no)`
		`FROM EMP`
		`GROUP BY Dept-no`
		`HAVING Salary > 1,00,000;`
			44. `SELECT student`
			`FROM FREQUENT`
			`WHERE parlor in ( SELECT parlor`
			`FROM SERVES`
			`WHERE ice-cream in ( SELECT ice-cream`
			`FROM LIKES`
			`WHERE`
			`LIKE.student = FR`	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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